IAN WHITEHOUSE

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EDUCATION

University of Maryland — **College Park, MD**

Doctor of Philosophy in Computer Science — *August 2024 – May 2029 (expected)*

- Won the National Defense Science and Engineering Graduate (NDSEG) Fellowship, a prestigious national fellowship
- Advised by Dr. William Regli and Dr. Wolfgang Losert, researching analog, neuromorphic, and living computing systems through dynamical systems and learning theory
- Relevant coursework: Computational Imaging, Machine Learning Theory, Numerical Optimization

American University — **Washington, DC**

Bachelor of Science in Computer Science, summa cum laude — *August 2020 – May 2024*

- Won the Lockheed Martin STEM Scholarship and Dean's Scholarship, inducted into Upsilon Pi Epsilon society
- Relevant Coursework: High-Performance Computing, Introduction to Simulation and Modeling, Differential Equations

SELECTED PUBLICATIONS **See all online**

K. Bhetwal, C. Mert, and **I. Whitehouse**, "Name-bias in large language models," 2026.

I. Whitehouse, A. Zenginoğlu, F. Klein, M. E. A. Maizer, W. Smith, S. Chan, S. Duddella, and W. Losert, "Quantum error correction and biological error correction: A structural analogy between qubits and neurons," arXiv preprint (under review), 2026.

I. Whitehouse, H. Kang, and W. Losert, "Emergent detection of concept drift within the glia-inspired 'rhythmic sharing' algorithm," npj Unconventional Computing, vol. 3, no. 1, 2026, Art. no. 25.

R. O'Loughlin, B. Oripov, N. Skuda, N. Chongsiriwatana, **I. Whitehouse**, W. Losert, B. Hayes, A. McCaughan, and S. Buckley, "$\vec{\delta}$ Multiplexed Gradient Descent: Perturbative learning with astrocytes," in *2026 Neuro Inspired Computational Elements (NICE)*, Atlanta, GA, USA, 2026, pp. 1–9.

I. Whitehouse, R. Yepez-Lopez, and R. Corizzo, "Distributed concept drift detection for efficient model adaptation with big data streams," in *2023 IEEE International Conference on Big Data*, Sorrento, Italy, 2023, pp. 5392–5397.

L. P. Damasceno, E. Rexhepi, A. Shafer, **I. Whitehouse**, N. Japkowicz, C. C. Cavalcante, R. Corizzo, and Z. Boukouvalas, "Exploiting sparsity and statistical dependence in multivariate data fusion: An application to misinformation detection for high-impact events," Machine Learning, vol. 113, no. 4, pp. 2183–2205, 2024.

EXPERIENCE

University of Maryland — **College Park, MD**

Research Assistant, Dr. William Regli's Group — *August 2024 – Present*

- Designed quantitative performance metrics for analog SAT solvers, adopted by 7 performers in a DARPA-funded program
- Benchmarked analog SAT solvers against state-of-the-art digital solvers on time-/energy-to-solution and pre-/post-processing overhead, finding that digital-to-analog conversion costs dominated the hardware's advantages
- Supervised five summer interns producing an experimental report and literature review on emerging computing substrates

Research Assistant, Dr. Wolfgang Losert's Group — *May 2025 – Present*

- Developed a glia-inspired learning algorithm for concept drift detection in evolving dynamical systems, exceeding prior state-of-the-art F1 by 84% on WADI water-utility anomaly detection benchmarks
- Built decoding and analysis pipelines for in-vitro living neuron reservoirs, in collaboration with biologists and physicists
- Characterized information propagation and spike-timing-dependent plasticity in living neuron reservoirs

U.S. Army Research Laboratory — **Adelphi, MD**

Neuromorphic Computing Intern — *May 2025 – August 2025*

- Adapted ARL neuromorphic algorithms for concept drift detection in collaboration with Lockheed Martin
- Built pipelines to extract effective connectivity via Granger causality from calcium recordings of living neurons

Leidos, Inc. — **Arlington, VA**

Machine Learning Intern, Leidos Innovation Center — *May 2023 – August 2024*

- Developed efficient, interpretable machine-learning models for low-SNR underwater acoustic sensing
- Used differential-equation models and digital signal processing to generate sensor data and analyze model behavior

American University — **Washington, DC**

Research Assistant, Institute for IDEAS and Department of Computer Science — *September 2021 – May 2024*

- Devised a novel machine-learning architectures for data fusion and concept drift detection, leading to 3 publications

Lockheed Martin **Syracuse, NY**
Student Software Engineer, Rotary and Mission Systems *May 2022 – August 2022*
• Wrote high-performance C++ software to improve parsing, modeling, and simulation of E-2D airborne radar systems

SELECTED PRESENTATIONS AND POSTERS ∗: Presenter **: See all online**

N. Chongsiriwatana∗, **I. Whitehouse**, A. Emenheiser, K. Perry, S. Gates III, L. Wijerathna, B. Babadi, and W. Losert, "Classification and information propagation in biological reservoir computers," ACM International Conference on Neuromorphic Systems (ICONS), Chicago, IL, 2026.

N. Skuda∗, R. O'Loughlin, B. Oripov, N. Chongsiriwatana, **I. Whitehouse**, W. Losert, B. Hayes, A. McCaughan, and S. Buckley, "δ Multiplexed Gradient Descent: Perturbative learning with astrocytes," IEEE Conference on Neuro-Inspired Computational Elements (NICE), Atlanta, GA, 2026.

I. Whitehouse∗, N. Chongsiriwatana, R. O'Loughlin, B. Oripov, N. Skuda, W. Losert, B. Hayes, A. McCaughan, and S. Buckley, "From astrocytes to novel algorithms: Multiplexed Gradient Descent and Rhythmic Sharing," IEEE Conference on Neuro-Inspired Computational Elements (NICE), Atlanta, GA, 2026.

I. Whitehouse∗, N. Chongsiriwatana, H. Kang, and W. Losert, "Rhythmic Sharing: An astrocyte-inspired algorithm for robust learning of evolving dynamical systems with applications in anomaly detection," Joint Mathematics Meeting (JMM), Washington, DC, 2026.

I. Whitehouse∗, H. Kang, and W. Losert, "Emergent detection of concept drift within the glia-inspired 'Rhythmic Sharing' algorithm," 10th International Conference on Rebooting Computing (ICRC), San Diego, CA, 2025.

I. Whitehouse∗ and G. Byrne, "Adversarial machine learning training for signal-to-noise generalization in passive undersea acoustics," Naval Applications of Machine Learning, San Diego, CA, 2024.

SKILLS AND INTERESTS

• **Scientific Computing**: Python (NumPy, JAX, PyTorch, TensorFlow, HuggingFace, Pandas), Apache Spark
• **Machine Learning Paradigms**: reservoir computing, convolutional networks, transformers, diffusion models
• **Software Engineering**: C++, Rust, Java, JavaScript, binary data ingestion
• **Professional Interests**: neuromorphic algorithms and hardware, living neuron computing, nonlinear dynamical systems